AYE

MECHANICAL ENGINEERING • LESSON 06

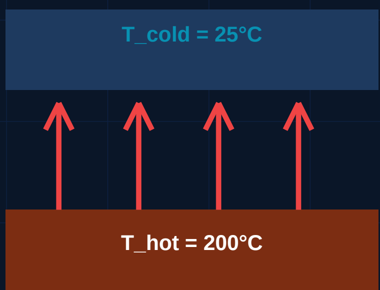
Heat Transfer

Conduction • Convection • Radiation

Fourier's Law | Newton's Cooling | Stefan-Boltzmann | Heat Exchangers

What You Will Learn

- Three modes: conduction, convection, and radiation — formulas and diagrams
- Fourier's Law and Newton's Law of Cooling with worked examples
- Thermal resistance concept and series/parallel thermal circuits
- Heat exchanger types, LMTD method, and counter vs parallel flow



Heat always flows from hot → cold

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MECH-006: Heat Transfer

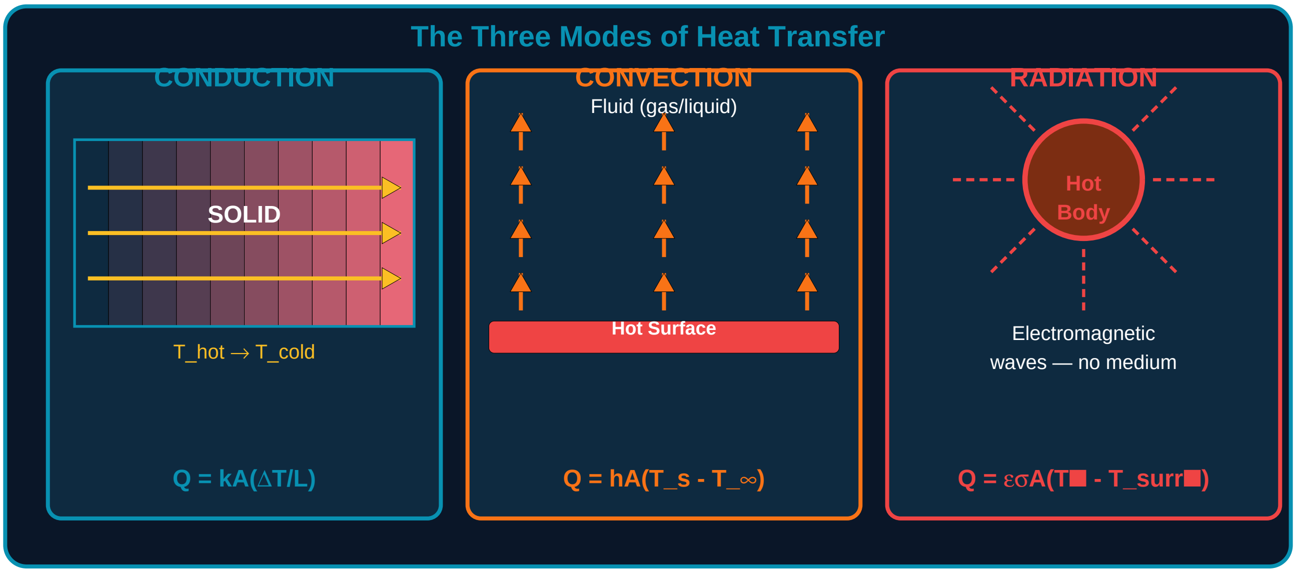
Heat transfer is the movement of thermal energy from a region of higher temperature to a region of lower temperature. It is one of the most fundamental topics in mechanical engineering, underpinning the design of engines, heat exchangers, HVAC systems, electronic cooling, power plants, and manufacturing processes. Every mechanical engineer must understand the three modes of heat transfer and be able to apply the governing equations to real engineering problems.

Learning Objectives

- Explain the three modes of heat transfer and identify which mode dominates in a given situation
- Apply Fourier's Law to calculate conduction heat flux through flat walls and cylindrical pipes
- Use Newton's Law of Cooling to solve convection problems with given heat transfer coefficients
- Calculate radiative heat transfer using the Stefan-Boltzmann law and emissivity values
- Build thermal resistance circuits and solve combined heat transfer problems
- Describe heat exchanger types and apply the LMTD method to find required surface area

1. The Three Modes of Heat Transfer

Heat transfer always occurs from a high-temperature body to a lower-temperature body. There are three distinct physical mechanisms — conduction, convection, and radiation — and in most real engineering problems, more than one mode acts simultaneously.

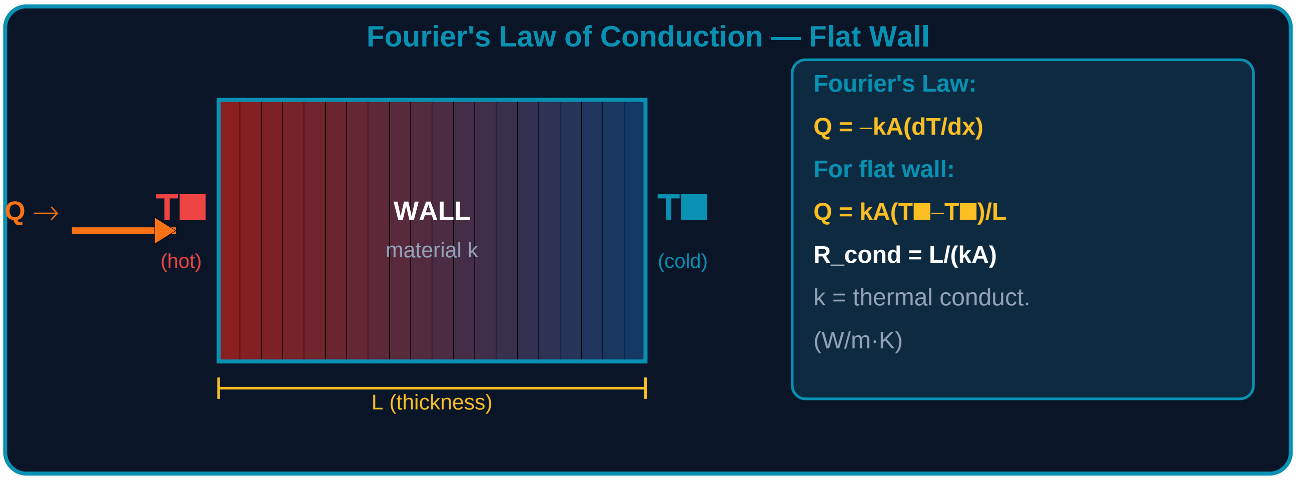


Mode	Medium Required	Governing Law	Typical Applications
Conduction	Solid or stationary fluid	Fourier's Law: $Q = kA(\Delta T/L)$	Walls, insulation, engine blocks, PCB cooling
Convection	Moving fluid (gas or liquid)	Newton's Cooling: $Q = hA(T_s - T_\infty)$	Radiators, heat sinks, HVAC ducts, boilers

Radiation	No medium — vacuum or transparent gas	Stefan-Boltzmann: Q = εσA(T■ – T_surr■)	Solar panels, furnaces, infrared heating, space thermal control

2. Conduction — Fourier's Law

Conduction is heat transfer through a solid or stationary fluid by direct molecular energy exchange. Faster-moving molecules collide with and transfer energy to slower-moving neighbours. No bulk movement of material occurs. The rate of conduction depends on the temperature gradient, the cross-sectional area, and the thermal conductivity of the material.



Fourier's Law — Key Variables

Symbol	Quantity	SI Unit	Notes
Q	Heat transfer rate	W (Watts)	Positive value = heat flowing in +x direction
k	Thermal conductivity	W/(m·K)	Property of material. High k = good conductor. Metals: 50–400. Insulation: 0.02–0.05
A	Cross-sectional area	m^2	Area perpendicular to heat flow direction
ΔT = T■ – T■	Temperature difference	K or °C	Driving force for conduction. Larger ΔT = more heat flow
L	Thickness of material	m	In direction of heat flow
R_cond = L/(kA)	Thermal resistance	K/W	Analogous to electrical resistance. Higher R = less heat flow

Thermal Conductivity of Common Engineering Materials

Material	k (W/m·K)	Category	Engineering Application
Pure copper	385	Excellent conductor	Heat sinks, electrical conductors, heat pipes
Aluminium alloy	155–220	Good conductor	Engine blocks, heat exchangers, electronics cooling
Carbon steel	50	Moderate conductor	Pressure vessels, pipes, structural components

Stainless steel	15–17	Moderate conductor	Food equipment, medical, corrosive environments
Concrete	0.8–1.4	Poor conductor	Building walls, foundations
Glass wool / fibreglass	0.03–0.05	Insulator	Building insulation, pipe lagging
Polyurethane foam	0.022–0.028	Excellent insulator	Refrigerators, cold stores, pipe insulation
Air (still)	0.026	Insulator	Double-glazing gap, vacuum flask
Water	0.6	Moderate	Liquid cooling systems, heat exchangers

3. Convection — Newton's Law of Cooling

Convection transfers heat between a solid surface and an adjacent moving fluid (liquid or gas). The fluid carries thermal energy away from (or towards) the surface by bulk motion. Convection is the dominant heat transfer mode in most HVAC, heat exchanger, and electronic cooling applications.

Type	Mechanism	h range ($W/m^2 \cdot K$)	Examples
Natural (free) convection	Buoyancy-driven flow. Hot fluid rises, cool fluid falls. No external forcing.	2–25 (air) 50–1000 (water)	Room heating by radiators, electronic component cooling in open air
Forced convection (gas)	External fan, pump, or wind drives fluid over surface.	25–250 (air)	Heat sinks with fans, air coolers, ventilation ducts
Forced convection (liquid)	Pump circulates liquid coolant over or through the heated surface.	100–20,000 (water)	Engine cooling jackets, liquid-cooled servers, chilled water coils
Boiling convection	Phase change at surface dramatically increases h due to latent heat.	2,000–100,000	Steam boilers, evaporators, water-cooled reactors
Condensing convection	Vapour condenses on cool surface, releasing latent heat.	5,000–100,000	Condensers, steam heating coils, refrigerant circuits

Convection Coefficient h — Key Formula

$$Q = h \times A \times (T_{surface} - T_{fluid})$$

The convection coefficient *h* is NOT a material property — it depends on the fluid, flow velocity, geometry, and surface condition. It must be determined experimentally or from Nusselt number correlations for the specific geometry.

4. Radiation — Stefan-Boltzmann Law

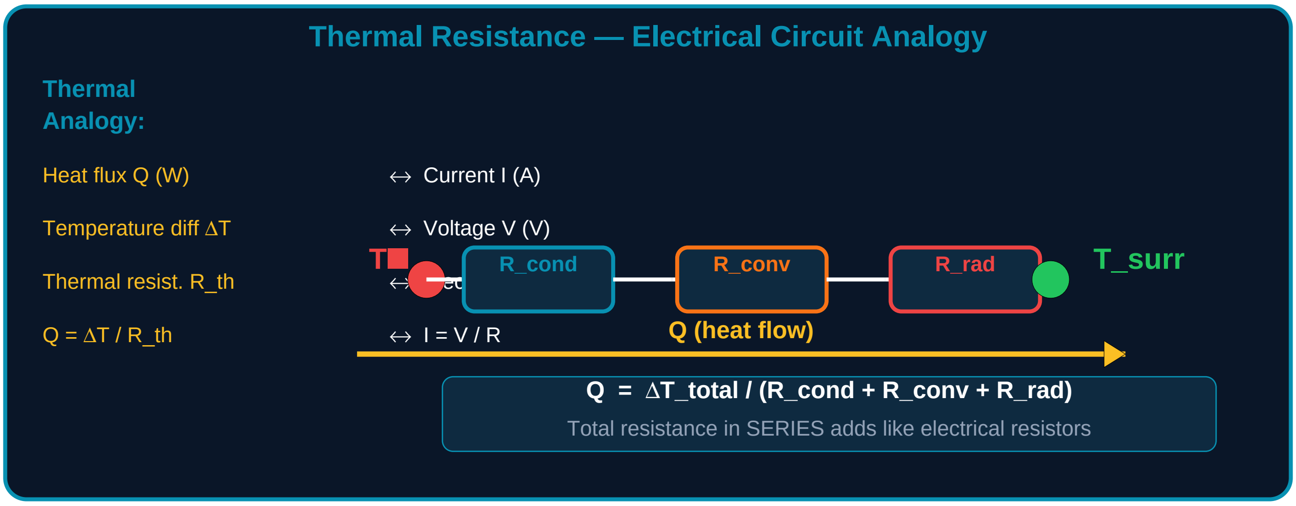
Radiation is the emission of electromagnetic energy from any body above absolute zero. Unlike conduction and convection, radiation requires no medium — it travels through a vacuum at the speed of light. At high temperatures (furnaces, combustion, solar), radiation becomes the dominant mode.

Symbol	Quantity	Value / Unit	Notes
Q	Net radiative heat transfer	W	Q = εσA(T■■ – T■■). Both temperatures MUST be in Kelvin.
ε (epsilon)	Emissivity	0 to 1 (dimensionless)	Blackbody = 1.0. Polished metal ≈ 0.02–0.05. Painted surface ≈ 0.85–0.95

σ (sigma)	Stefan-Boltzmann constant	5.67 × 10■■ W/(m²·K■)	Universal physical constant — same for all materials
A	Surface area	m²	Total area radiating energy
T (K)	Absolute temperature	Kelvin (K = °C + 273.15)	CRITICAL: Always convert to Kelvin before calculating T■
View factor F	Fraction of radiation leaving surface 1 that reaches surface 2	0 to 1	For large parallel surfaces facing each other: F ≈ 1. Complex geometry requires calculation.

5. Thermal Resistance and Thermal Circuits

The thermal resistance concept is the most powerful tool for solving combined heat transfer problems. Every mode of heat transfer can be expressed as a resistance: $R = \Delta T/Q$. Multiple resistances in series add like electrical resistors in series, enabling complex multi-layer wall problems to be solved by simple algebra.



Thermal Resistance Formulas

Mode	Resistance Formula	Units	When to Apply
Conduction (flat wall)	R = L / (k × A)	K/W	Each solid layer in a composite wall (insulation + brick + plaster)
Conduction (cylinder)	R = ln(r■/r■) / (2πkL)	K/W	Pipe insulation, cylindrical pressure vessels
Convection	R = 1 / (h × A)	K/W	At every fluid-solid interface. One for inner surface, one for outer surface of a wall
Radiation	R = 1 / (εσA(T■²+T■²)(T■+T■))	K/W	Linearised for small temperature differences. Exact at high temperatures using Q = εσA(T■■–T■■)
Contact resistance	R_c = R_contact / A	K/W	At material interfaces — poorly joined surfaces trap air, creating a thin high-resistance layer

Worked Example — Composite Wall

Problem: A factory wall has three layers: 150 mm brick (k=0.72 W/m·K), 50 mm fibreglass insulation (k=0.038 W/m·K), and 12 mm plasterboard (k=0.16 W/m·K). Inside temperature 22°C, outside -5°C. Convection coefficients: inside h=10 W/m²·K, outside h=25 W/m²·K. Find heat loss per m² of wall.

Layer	Formula	Calculation	R (K/W per m²)
Inside convection	R = 1/h	1 / 10	0.100
Brick (150 mm)	R = L/k	0.150 / 0.72	0.208

Fibreglass (50 mm)	R = L/k	0.050 / 0.038	1.316
Plasterboard (12 mm)	R = L/k	0.012 / 0.16	0.075
Outside convection	R = 1/h	1 / 25	0.040
TOTAL R_total	—	Sum of all layers	1.739 K/W per m²
Heat flux Q/A	Q = ΔT/R_total	(22–(–5)) / 1.739	15.5 W/m²

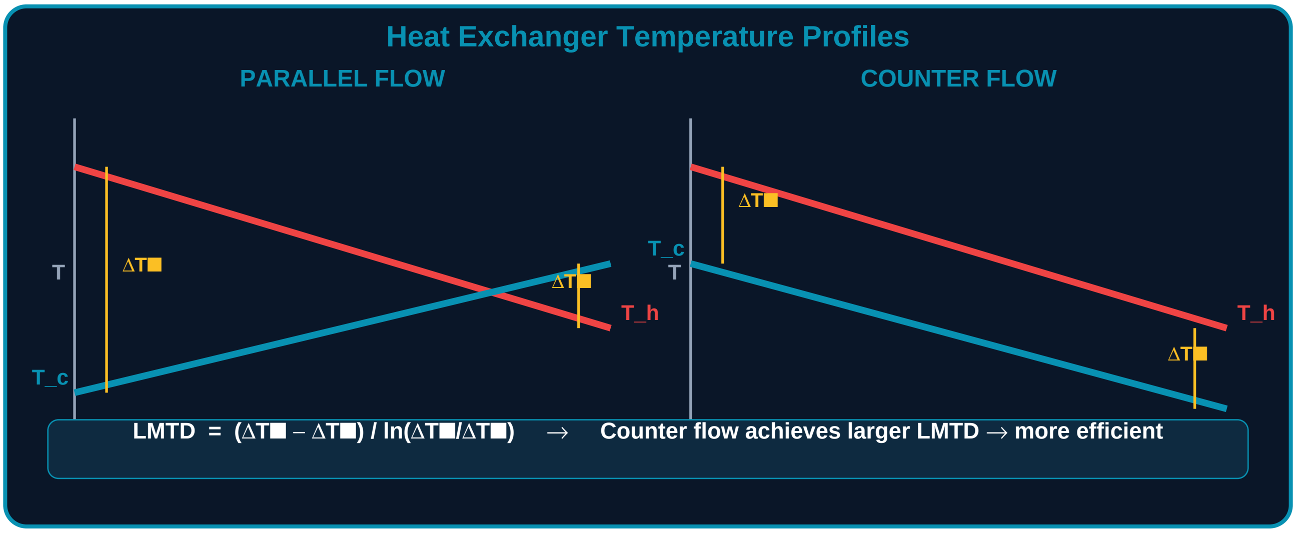
6. Heat Exchangers

A heat exchanger transfers heat between two fluids without mixing them. They are fundamental components in power plants, HVAC chillers, engine cooling systems, refrigeration circuits, chemical processes, and oil refining. Design requires calculating required heat transfer area from the Log Mean Temperature Difference (LMTD) method.

Types of Heat Exchangers

Type	How It Works	Advantages	Typical Applications
Shell and tube	One fluid flows through tubes, other around the shell. Usually multi-pass.	High pressure rating, large capacity, easy to clean	Power plant condensers, oil coolers, process industry
Plate heat exchanger	Corrugated metal plates pressed together. Fluids alternate between plates.	Very high h due to turbulence, compact, easy to expand capacity	HVAC chiller barrels, food processing, district heating
Double pipe (hairpin)	One tube inside another. Simplest design.	Simple, low cost, easy to analyse	Small capacity, laboratory, pilot plants
Cross-flow	One fluid flows perpendicular to the other.	Compact, low pressure drop on gas side	Air-to-water coils, car radiators, air-handling units
Condensers / evaporators	Phase change in one or both fluids.	Very high h due to latent heat	Refrigeration, steam power cycles, distillation

Temperature Profiles and LMTD Method



LMTD Method Step	Formula / Action
1. Find ΔT■ and ΔT■	ΔT■ = temperature difference at one end of HX. ΔT■ = difference at other end. For counter flow: ΔT■ = T_h,in – T_c,out, ΔT■ = T_h,out – T_c,in
2. Calculate LMTD	LMTD = (ΔT■ – ΔT■) / ln(ΔT■ / ΔT■). Use L'Hopital rule: if ΔT■ ≈ ΔT■, LMTD ≈ ΔT■
3. Find required area	Q = U × A × LMTD → A = Q / (U × LMTD). U = overall heat transfer coefficient (W/m²·K) combining both fluid convection and wall conduction.

4. Apply correction factor F	Q = F × U × A × LMTD. F accounts for non-ideal flow in multi-pass or cross-flow exchangers. F from TEMA charts (typically 0.8–0.98).

7. Practice Exercises

#	Problem	Given Data / Steps
1	Conduction through insulated pipe	Steam pipe: k_steel=50, wall thickness=6 mm, k_insulation=0.04, insulation 50 mm thick. T_steam=180°C, T_ambient=25°C. Find heat loss per metre of pipe length.
2	Newton's Law of Cooling	A metal block (surface area 0.08 m^2) at 120°C is cooled by forced air. h=55 $W/m^2 \cdot K$, T_air=25°C. Calculate heat removal rate Q in watts.
3	Radiation from a furnace wall	Furnace wall surface: ε=0.85, area=4 m^2, T_wall=600°C, T_surroundings=20°C. Calculate net radiative heat loss. (Remember: K = °C + 273.15)
4	Composite wall thermal circuit	Wall: inside convection h=8, brick 200 mm k=0.75, polystyrene 80 mm k=0.035, outside convection h=20. T_in=20°C, T_out=–10°C. Find Q/m^2 and temperature at each interface.
5	Counter-flow heat exchanger	Oil (C_p=2 kJ/kg·K, ■=2 kg/s) cooled from 90°C to 50°C by water entering at 20°C, leaving at 45°C. U=350 $W/m^2 \cdot K$. Find required surface area using LMTD method.

Lesson Summary

Topic	Key Formula	Critical Note
Conduction	Q = kA(T■–T■)/L R_cond = L/(kA)	k is a material property. Higher k = better conductor.
Convection	Q = hA(T_s – T_∞) R_conv = 1/(hA)	h depends on flow conditions, NOT just the fluid. Must be measured or correlated.
Radiation	Q = εσA(T■■ – T■■)	MUST use Kelvin. σ = 5.67 × 10■■ W/m²K■. ε = 1 for blackbody.
Thermal resistance	R_total = R■ + R■ + ... (series)	Analogy: heat flow = voltage, Q = current, R = resistance.
LMTD method	Q = U × A × LMTD	Counter-flow gives larger LMTD than parallel flow → smaller area needed.
Conduction (cylinder)	R = ln(r■/r■) / (2πkL)	Used for pipe insulation calculations — very common in HVAC and piping.

What Is Next — MECH-007

MECH-007: Machine Elements — Shafts, keys, couplings, bearings, gears, belts and chains, springs, and fasteners. How to select and size each element for load, speed, and life requirements. Design factor and fatigue loading concepts. Essential knowledge for any mechanical or production engineer working on machinery.

Resource	Details
PDF Series	MECH-001 through MECH-030 at ayetechub.com/pdfs.html

Heat Transfer Calculator	Engineering Formulas Reference — ayetechub.com/downloads
Community	AYE Tech Hub Telegram group: t.me/ayetechub